Benchmarking Quantum Gates and Circuits

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Comprehensive Summary with Expert Intuition

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Algorithm/Protocol **Pros** **Cons** **Expert Intuition** **Key result** **Math detail**

1 Introduction and Motivation

- Fault-tolerant quantum computing requires gate fidelities exceeding accuracy thresholds ($p_{\text{err}} < 10^{-4}$ for chemical accuracy, 10^{-5}–10^{-6} for dynamics)
- **Gate errors** are a fundamental bottleneck — each extra order of magnitude in fidelity ⇒ disproportionately lower qubit counts / circuit depths
- This paper: review of 6 major benchmarking methods + introduction of Deterministic Benchmarking (DB), a novel protocol
- Part of the broader field of **QCVV** (Quantum Characterization, Verification & Validation)

- **Why benchmarking matters for chemistry**: VQE/QPE require gate decompositions that preserve coherence over large circuit depths; any fidelity deviation biases energy predictions
- **Coherent vs. incoherent**: Coherent errors add as *amplitudes* (quadratically worse), incoherent add as *probabilities*. Traditional RB deliberately randomizes noise ⇒ can miss coherent errors
- **Philosophy**: No single benchmarking method is universally best — the trade-off is always *comprehensiveness vs. resource cost vs. simplicity*
- Not covered in this review: cycle benchmarking, quantum volume, mirror benchmarking, ML-based benchmarking, EPLG, etc. (these are mostly variants of the core methods reviewed here)

2 Gate Errors

2.1 Quantum Dynamical Maps

- A quantum process (channel) $\mathcal{E}$ is a **CPTP map**: $\mathcal{E}(\rho) = \sum_i A_i \rho A_i^\dagger$, with $\sum_i A_i^\dagger A_i = I$
- **Process matrix** χ: Expand Kraus operators in a fixed Hermitian basis $\{E_m\}_{m=0}^{D^2-1}$ satisfying $\mathrm{Tr}(E_m^\dagger E_n) = D\,\delta_{mn}$:

$$\mathcal{E}(\rho) = \sum_{m,n=0}^{D^2-1} \chi_{mn}\, E_m\, \rho\, E_n^\dagger, \qquad \chi_{mn} = \sum_i a_{in}^* a_{im} = \chi_{nm}^* \tag{1}$$

  χ is Hermitian, positive semidefinite; encodes $D^4 - D^2$ real parameters (12 for a single qubit)
- **Choi–Jamiołkowski isomorphism**: Map $\mathcal{E} \leftrightarrow$ Choi state $\rho_{\mathcal{E}} = (I \otimes \mathcal{E})(|\phi\rangle\langle\phi|)$ where $|\phi\rangle = \frac{1}{\sqrt{D}} \sum_i |i\rangle \otimes |i\rangle$
- **Process fidelity**: $F_{\mathrm{proc}}(\mathcal{E}, \mathcal{E}') = F(\rho_{\mathcal{E}}, \rho_{\mathcal{E}'}) = \|\sqrt{\rho_{\mathcal{E}}}\sqrt{\rho_{\mathcal{E}'}}\|_1$ (Uhlmann fidelity between Choi states)

2.2 Coherent vs. Incoherent Errors

Coherent Errors	Incoherent Errors
Purity preserved: $\mathrm{Tr}(\rho^2) = 1$	Purity lost: $\mathrm{Tr}(\rho^2) < 1$
Deterministic, repeatable	Random (stochastic)
Amplitudes add in phase $\Rightarrow$ failure prob. grows quadratically	Probabilities add $\Rightarrow$ failure prob. grows linearly
Always reversible in principle	Requires QEC to reverse
More damaging in general	

2.3 Single-Qubit Gate Errors

Bloch vector $\vec{v} = (\langle\sigma_x\rangle, \langle\sigma_y\rangle, \langle\sigma_z\rangle)$; general gate $R(\vartheta, \theta, \phi)$ = rotation by ϑ about axis (θ, ϕ).

Coherent errors:

- **Rotation errors**: ϑ deviates from target (drive amplitude drifts, coupling errors)
- **Phase errors**: rotation axis altered (θ and/or ϕ change); from qubit frequency drifts, control phase drifts
- **Leakage**: state evolves outside qubit space (e.g., transmon higher levels); coherent if it preserves purity

Incoherent errors:

- **Relaxation (T_1)**: bit-flip errors, exponential decay toward equilibrium; nonzero purity at finite T
- **Dephasing (T_2, T_ϕ)**: phase-flip errors; Markovian + $1/f$ noise components common in solid-state; $1/T_2 = 1/(2T_1) + 1/T_\phi$
- **Erasure**: special leakage where qubit info lost with a known signature; easier to correct than general errors

Depolarizing model (lumps incoherent errors):

$$\mathcal{E}(\rho) = (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z) \tag{2}$$

Limitation: obscures physical origin, ignores biased noise & finite temperature.

Total parameter count (Markovian, ignoring erasure): 7 parameters — $\delta\vartheta$, $\delta\theta$, $\delta\phi$, γ_1 (relaxation), equilibrium population, γ_ϕ (dephasing), leakage angle.

2.4 Two-Qubit Gate Errors

- Error landscape is *much richer* than single-qubit: device/implementation specific
- Example (parametrically-activated iSWAP in transmons): errors include T_1/T_2 of each qubit, $1/f$ dephasing, coupler noise, incomplete swapping, ZZ interaction, phase errors, leakage
- **Best practice**: develop an "error budget"; full QPT once, then targeted measurements
- **Key danger**: amplification of single-qubit coherent errors when amplifying two-qubit errors — use dynamical decoupling between gate applications
- Two-qubit incoherent errors (joint bath couplings) are often ignored but can be significant (e.g., coupler as joint bath)

3 Gate and Circuit Benchmarking Protocols

Criteria for judging a benchmarking protocol:

1. **Comprehensiveness**: How much error information? (number of parameters characterized)
2. **Specificity**: What noise model assumptions? (gate-independent? Pauli? etc.)
3. **Simplicity**: How easily adapted to existing setups?
4. **Resource cost**: How expensive to run? (number of circuits, measurements, classical post-processing)
5. **Robustness to SPAM**: Critical requirement for most protocols

3.1 Randomized Benchmarking (RB)

3.1.1 Standard RB

1. **Prepare** $|0\rangle^{\otimes n}$
2. **Apply** m random Clifford gates $\{C_i\}_{i=1}^m$: $\quad U_m = C_m C_{m-1} \cdots C_1$
3. **Inversion gate**: $C_{\text{inv}} = C_{m+1} = U_m^\dagger = C_1^\dagger \cdots C_m^\dagger$ (efficiently computable classically)
4. **Measure** in computational basis; record if system returned to $|0\rangle^{\otimes n}$
5. **Average** over K random sequences at each depth $m \Rightarrow$ survival probability $P(m)$
6. **Fit**: $P_{\text{avg}}(m) = A\,p^m + B$ to extract depolarization parameter p

Groups: **Pauli group** $\mathcal{P}_n$ = all n-fold tensor products of $\{I, X, Y, Z\}$. **Clifford group** = gates C such that $CPC^\dagger \in \mathcal{P}_n$ for all $P \in \mathcal{P}_n$ (preserves Pauli group under conjugation).

Key idea: Random Clifford gates *twirl* the error map into an effective **depolarizing channel**:

$$\mathcal{E}_{\text{dep}}(\rho) = p\,\rho + (1-p)\frac{I}{D} \tag{3}$$

RB Theory: Assuming gate-independent noise $\mathcal{E}_i \equiv \mathcal{E}$:

$$P_{\text{avg}}(m) = \text{Tr}\left[\rho_0\,\mathcal{E}\left(p^m \rho_0 + (1-p^m)\frac{I}{D}\right)\right] = A\,p^m + B \tag{4}$$

where $A = \text{Tr}\left[\rho_0\,\mathcal{E}\left(\rho_0 - \frac{I}{D}\right)\right]$, $B = \text{Tr}\left[\rho_0\,\mathcal{E}\left(\frac{I}{D}\right)\right]$.
A and B capture SPAM errors. Average Clifford gate infidelity:

$$\boxed{r_C = \frac{D-1}{D}(1-p)} \tag{5}$$

Gate-dependent noise correction (1st order): $P_{\text{avg}}^{(1)}(m) = A_1 p^m + B_1 + C_1(m-1)(q-p^2)p^{m-2}$; $(q-p^2)$ measures gate-dependence.

3.1.2 Interleaved RB (IRB)

1. Run **standard RB** $\Rightarrow$ extract decay parameter p
2. Run **interleaved sequence**: insert target gate G between every pair of random Cliffords:

$$U_{m,\text{inter}} = G_m \circ G \circ G_{m-1} \circ \cdots \circ G \circ G_2 \circ G \circ G_1$$

   followed by $G_{m+1} = U_{m,\text{inter}}^{-1} \Rightarrow$ extract p_{inter}
3. Gate infidelity of G: $r_G = \frac{D-1}{D}\left(1 - \frac{p_{\text{inter}}}{p}\right)$

Caveat: Coherent errors in G can *add or cancel* with random Clifford errors $\Rightarrow$ possibly **negative** r_G (unphysical $F > 1$). Error bound: $r_G \in [r_G - E,\, r_G + E]$ with

$$E = \min\left\{\frac{(D-1)|p - p_{\text{inter}}/p| + (1-p)}{D},\ \frac{2(D^2-1)(1-p)}{pD^2} + \frac{4\sqrt{1-p}\,\sqrt{D^2-1}}{pD^2}\right\}$$

Extensions of RB: Simultaneous RB, Leakage RB, Non-Markovian RB, Mirror RB, Character RB (beyond Clifford group).

- SPAM-robust (captured by A, B)
- Simple to implement; scalable
- Single-number summary useful for quick comparison
- Widely adopted, well understood theory

- Reduces error to a *single number* — loses all specificity
- Insensitive to coherent errors (twirling converts them to depolarizing)
- Assumes gate-independent, time-independent noise
- Gate-dependent corrections require large m conflicting with validity conditions

- RB is the "quick health check" of quantum gates — tells you the *average* error rate, but not *what* is wrong
- The twirling trick is powerful but also RB's main weakness: by converting all noise to depolarizing, coherent errors that may accumulate quadratically in circuits are rendered invisible
- In practice, RB fidelity can look great while the actual circuit performance is poor because coherent error dominates
- IRB is unreliable when $p_{\text{inter}} > p$; always check the error bound E

3.2 Quantum Process Tomography (QPT)

3.2.1 Standard QPT (SQPT)

1. Prepare D^2 linearly independent input states $\{\rho_k\}$
2. Apply the map $\mathcal{E}$ to each
3. Perform quantum state tomography on each output $\mathcal{E}(\rho_k)$: measure expectations of D^2 basis operators $\{E_k\}$
4. **Total measurements**: D^4
5. Reconstruct χ via matrix inversion: $B\chi = \lambda$, where B is $D^4 \times D^4$, λ contains tomographic data

Key equations: Express $\mathcal{E}(\rho_k) = \sum_l \lambda_{kl}\, \rho_l$ with $\lambda_{kl} = \mathrm{Tr}(\rho_k\, \mathcal{E}(\rho_l))$. Expand $E_m \rho_k E_n^\dagger = \sum_l B_{mn,lk}\, \rho_l$. Then $\sum_{mn} B_{mn,lk}\, \chi_{mn} = \lambda_{kl}$.

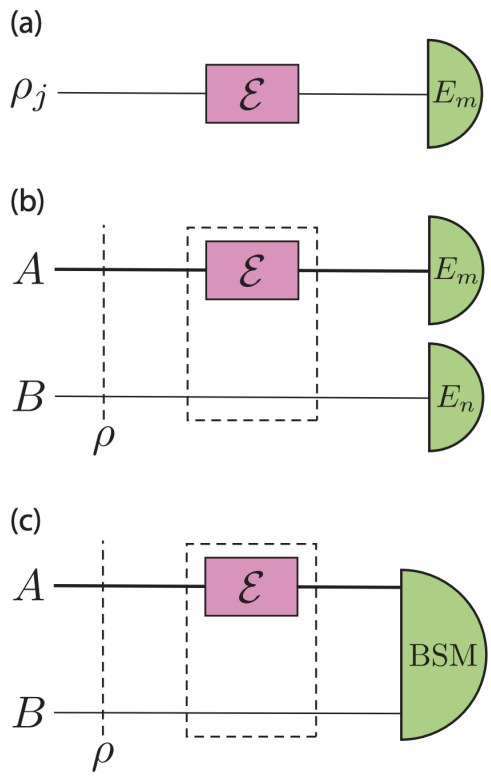

Figure 1: Three QPT variants: (a) SQPT, (b) Separable AAPT, (c) DCQD. Note DCQD uses Bell-state measurement.

3.2.2 Ancilla-Assisted Process Tomography (AAPT)

- Based on **Choi–Jamiołkowski isomorphism**: QPT $\equiv$ state tomography of the Choi state
- Attach ancilla B; prepare "faithful" joint state ρ_{AB} (must have maximal Schmidt number D^2)
- Apply $\mathcal{E} \otimes I$ on the joint system; perform state tomography on output
- Reconstruct χ via **two matrix inversions**: $\tilde{\alpha} = \tilde{\chi}\, \varrho$ then invert to find χ
- Maximally entangled states are *optimal* (minimize experimental error amplification from matrix inversion)
- Separable vs. non-separable measurements: non-separable offers no practical advantage (requires many-body interactions)

3.2.3 Direct Characterization of Quantum Dynamics (DCQD)

1. **Diagonal elements** χ_{mm}: Prepare maximally entangled $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$; apply $\mathcal{E} \otimes I$; perform Bell-state measurement (BSM):
$$p_m = \mathrm{Tr}[P_m\, (\mathcal{E} \otimes I)(|\Phi^+\rangle\langle\Phi^+|)] = \chi_{mm}$$
2. **Off-diagonal elements** $\chi_{m\neq n}$: Prepare non-maximally entangled $|\Phi_\alpha^+\rangle = \alpha|00\rangle + \beta|11\rangle$; BSM gives χ_{03}, χ_{12}. Basis changes via $H^A H^B$ and $S^A S^B H^A H^B$ give remaining elements.
3. **Total**: 4 input states, same BSM apparatus $\Rightarrow$ complete χ

Key advantage: No matrix inversion needed — measurement outcomes *directly* encode χ elements.
Scaling: DCQD reduces experimental configurations from D^{4n} (SQPT) to D^{2n} for n qudits.

Input state	Stabilizer	BSM equivalent	Output
$\lvert\Phi^+\rangle$	$Z^A Z^B$, $X^A X^B$	$P_{\Psi\pm}$, $P_{\Phi\pm}$	$\chi_{00}, \chi_{11}, \chi_{22}, \chi_{33}$
$\lvert\Phi^+_\alpha\rangle$	$Z^A Z^B$	$P_{\Phi+} \pm P_{\Phi-}$, $P_{\Psi+} \pm P_{\Psi-}$	χ_{03}, χ_{12}
$\lvert\Phi^+_\alpha\rangle_X$	$X^A X^B$	$P_{\Phi+} \pm P_{\Psi+}$, $P_{\Phi-} \pm P_{\Psi-}$	χ_{01}, χ_{23}
$\lvert\Phi^+_\alpha\rangle_Y$	$Y^A Y^B$	$P_{\Phi+} \pm P_{\Psi-}$, $P_{\Phi-} \pm P_{\Psi+}$	χ_{02}, χ_{13}

- **Full characterization** of the process ($D^4 - D^2$ params)
- DCQD: quadratic reduction in experiments
- DCQD: no inversion needed, single measurement apparatus

- **SQPT**: D^4 measurements, scales very poorly
- **SQPT**: sensitive to SPAM errors
- **AAPT**: requires ancilla + entanglement
- **DCQD**: assumes negligible SPAM (can be partially relaxed)

- QPT is the "gold standard" for complete characterization but *impractical beyond a few qubits*
- DCQD's use of **stabilizer codes** / error-detection ideas is elegant: the Bell state $\lvert\Phi^+\rangle$ is stabilized by $Z^A Z^B$ and $X^A X^B$; an error E anticommuting with a stabilizer is detected by measuring it
- The Choi–Jamiołkowski isomorphism is the conceptual backbone of many methods: it converts process characterization to state characterization

3.3 Gate Set Tomography (GST)

1. **Define gate set**: $\mathcal{G} = \{|\rho^{(j)}\rangle\rangle, \{G_j\}, \langle\langle M_j|\}$ (state preparations, gates, measurements)
2. **Define fiducial gates** $\{F_i\}$ from the gate set to form informationally-complete state preparation & measurement: $|\rho^{(j)}\rangle\rangle \to F_j|\rho^{(j)}\rangle\rangle$, $\langle\langle M_i| \to \langle\langle M_i|F_i$
3. **Run circuits**: $F_i\, G_k\, F_j$ on initial state ρ; measure M_i (or $F_i\, G_k^l\, F_j$ for long-sequence GST)
4. **Record** outcome probabilities m_{ijk} (average over ~ 2000 repetitions per setting)
5. **Maximum Likelihood Estimation**: Parameterize $\chi_G = T^\dagger T$ (Cholesky); minimize

$$L(\mathcal{G}) = \sum_{ijk} \frac{(m_{ijk} - p_{ijk}(\vec{\theta}))^2}{\sigma_{ijk}^2}, \qquad p_{ijk}(\vec{\theta}) = \langle\langle M_i(\vec{\theta})|\, F_i(\vec{\theta})\, G_k(\vec{\theta})\, F_j(\vec{\theta})\, |\rho(\vec{\theta})\rangle\rangle$$

6. **Gauge freedom**: Outcomes invariant under S: $G \to SGS^{-1}$, $|\rho\rangle\rangle \to S|\rho\rangle\rangle$, $\langle\langle M| \to \langle\langle M|S^{-1}$. Choose S maximizing fidelity.

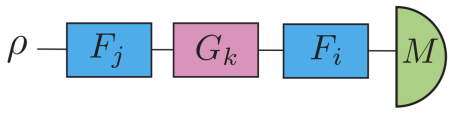

Figure 2: GST circuit: fiducials F_i, F_j, native gate G_k, measurement M on initial state ρ.

Process matrix parameterization: χ_G has $(D^4 - D^2)/2$ real free params (via Cholesky $\chi_G = T^\dagger T$ + trace-preservation: $\sum_{ij} \chi_{ij} \mathrm{Tr}(E_i E_k E_j) = \delta_{k0}$, giving D^2 constraints).

Example gate set: $\{G\} = \{I, \sqrt{X}, \sqrt{Y}\}$, fiducials $\{F\} = \{I, \sqrt{X}, \sqrt{Y}, \sqrt{X} \circ \sqrt{X}\}$. Uses $\sqrt{X} \circ \sqrt{X} = X$ instead of X to reduce cost.

Long-sequence GST: Apply $F_i\, G_k^l\, F_j$ (repeated l times) to *amplify* coherent errors.

Resource cost: 80 experiments for 1 qubit; > 4000 for 2 qubits; each averaged ~ 2000 times.

- **SPAM-robust** (models SPAM explicitly)
- Full gate set characterization (comprehensive as QPT)
- Amplifies coherent errors (long-sequence variant)
- Self-consistent: no prior calibration needed

- **Extremely resource-intensive**: 80+ experiments (1Q), 4000+ (2Q)
- Cannot capture non-Markovian noise
- Over-complete: much of the data is redundant
- Complex classical post-processing (MLE optimization)

- GST is the "full picture with SPAM included" — it's the most complete method but also the most expensive
- The gauge freedom is subtle: physically measurable quantities must be gauge-invariant, but the *representation* of the gate set depends on the gauge choice
- Key conceptual point: **fiducial gates serve as an internal reference frame**, eliminating the need for pre-calibrated states or measurements
- Recent work has reduced 2-qubit experiments by $\sim 8\times$, but still impractical for many-qubit systems

3.4 Direct Fidelity Estimation (DFE)

1. Define ideal gate U and noisy implementation $\tilde{U}$; their Choi states are ρ_U and $\rho_{\tilde{U}}$
2. **Average output fidelity**: $\bar{F}(U, \tilde{U}) = \frac{D\, F_{\text{proc}}(\rho_U, \rho_{\tilde{U}}) + 1}{D+1}$
3. Compute directly (no entangled state $|\phi\rangle$ needed):

$$F_{\text{proc}}(\rho_U, \rho_{\tilde{U}}) = \frac{1}{D^4} \sum_{j,k} \text{Tr}(U(P_j)P_k)\ \text{Tr}(\tilde{U}(P_j)P_k)$$

4. Determine tuples (P_j, P_k) for which $\text{Tr}(U(P_j)P_k) \neq 0$; call this set $\mathcal{A}$
5. **Importance sampling**: sample from $\mathcal{A}$ with probability $\mathcal{P}(P_j, P_k) = \text{Tr}(U(P_j)P_k)^2/D^4$
6. For each sample: prepare D eigenstates of P_j, apply $\tilde{U}$, measure P_k
7. Average N samples $\Rightarrow$ fidelity estimate $\bar{F}$

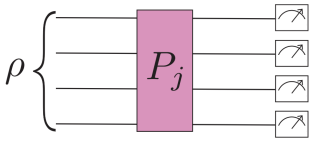

Figure 3: DFE: Estimate state fidelity by sampling random Pauli expectations.

Underlying Monte Carlo: Fidelity $F(\sigma, \rho) = \text{Tr}(\sigma\rho) = \sum_k p_k x_k = \langle X \rangle$ where $x_k = \rho_k/\sigma_k$, $p_k = \sigma_k^2/D$.
Sample complexity: Worst case exponential in n; polynomial for most states of interest. Demonstrated on 7 qubits.

- No assumptions on noise model
- Efficient for most practical states
- Only needs local preparations + measurements
- No entangled state preparation required

- **Susceptible to SPAM errors**
- Worst-case exponential sample complexity
- Provides only a single fidelity number
- Limited adaptation for routine benchmarking

3.5 Process Fidelity Estimation (PFE)

1. Consider ideal unitary U with projective measurement Π: ideal map $\Phi = \Pi \circ U$, noisy $\tilde{\Phi} = \Pi \circ \tilde{U}$
2. Define noise map $\Lambda_U = \tilde{U} \circ U^\dagger$, so $\tilde{U} = \Lambda_U \circ U$
3. **Key result**:

$$\boxed{F_{\text{proc}}(\Phi, \tilde{\Phi}) = \frac{1}{D} \sum_{k=1}^{D} \sqrt{\Pr(k \mid \Lambda_U(\pi_k))} = \frac{1}{D} \sum_{k=1}^{D} \sqrt{\langle k | \tilde{U}(U^\dagger(\pi_k)) | k \rangle}}$$

4. **Operational meaning**: Prepare time-reversed states $|\psi_k\rangle = U^\dagger |k\rangle$, then apply $\tilde{U}$, measure prob. of outcome k
5. **Unbiased estimator** (via sampling m values of k):

$$F_{\text{proc}} \approx \frac{m}{m-1} \left(\frac{1}{m} \sum_{l=1}^{m} \sqrt{\Pr(k_l | \Lambda_U(\pi_{k_l}))} \right)^2 - \frac{1}{m(m-1)} \sum_{l=1}^{m} \Pr(k_l | \Lambda_U(\pi_{k_l}))$$

- Generalizes DFE to include measurements (applicable to entire algorithms)
- No entangled state preparation needed
- Unitarily invariant fidelity

- Requires preparing time-reversed states $U^\dagger |k\rangle$ with high fidelity
- Exponentially many initial states (must sample)
- $O(1/m)$ bias in naive estimator

- PFE's conceptual innovation: by including the measurement in the benchmark, it captures *end-to-end* circuit performance
- The key equation derives from Choi state analysis + the fact that $\rho_\Pi^2 = \rho_\Pi / D$ (projectors on computational basis are pure)
- The time-reversed state preparation $U^\dagger |k\rangle$ can sometimes be done with only Hadamard + virtual phase gates

3.6 Cross-Entropy Benchmarking (XEB)

1. **Sample** random circuit C_i from ensemble RCS(n, P): P layers of 2-qubit gates interleaved with Haar-random 1-qubit gates
2. **Run** noisy circuit $\tilde{C}_i$ on quantum hardware; collect M bitstring samples $\{x_i\}_{i=1}^{M}$
3. **Classically simulate** ideal circuit to get $p_U(x_i)$ for each sampled bitstring
4. **Estimate fidelity** via linear cross-entropy:

$$\boxed{\hat{F}_{\text{XEB}} = \frac{D}{M}\sum_{i=1}^{M} p_U(x_i) - 1}$$

5. **Repeat** for L random circuits; average: $\hat{F}_P = \frac{1}{L}\sum_i \hat{F}_{P,i}$
6. **Repeat** for varying depths $P \in \{1, \ldots, P_{\max}\}$
7. **Fit**: $\hat{F}_P = A\, e^{-\lambda P}$, extract noise strength $\lambda = \sum_\alpha p_\alpha$ (Pauli error rates)

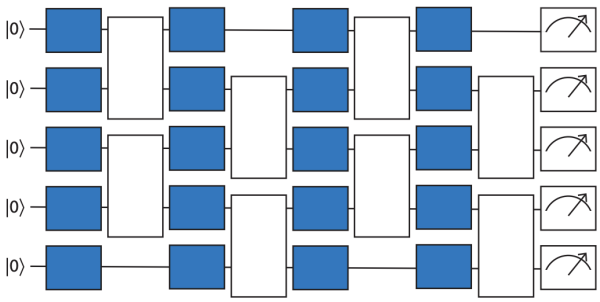

Figure 4: RCS benchmarking circuit: layers of 2-qubit gates (white) interleaved with Haar-random 1-qubit gates (blue), followed by measurement.

Noise model: $\rho_{\tilde{U}} = F_U|\psi_U\rangle\langle\psi_U| + (1-F_U)\sigma_U$. **Central assumption**: for deep random circuits, $\mathbb{E}_U[p_{\sigma_U}] \approx$ uniform distribution (concentration of measure).

Unbiased estimator: $\hat{F}_{\text{uXEB}} = \frac{(D/M)\sum_i p_U(x_i) - 1}{D\sum_x p_U(x)^2 - 1}$ (denominator $\to 1$ at log depth for random circuits).

Variance: $\text{Var}(\hat{F}_{\text{uXEB}}) = O(1/M) + \lambda^2(\mathbb{E}[F])^2 \Rightarrow$ needs $M = O(1/\epsilon^2)$ samples per circuit.

- Scales to ∼50 qubits (limited by classical simulation)
- SPAM-robust (depth-independent fit params)
- **Sample-efficient**: $O(1/\epsilon^2)$ measurements per circuit
- Works with arbitrary non-Clifford gates
- Basis for quantum supremacy experiments

- Requires **classical simulation** of ideal circuit (infeasible for very large circuits)
- Insensitive to I/Z errors (don't affect computational basis measurements)
- Not a good fidelity proxy for time-correlated errors
- Strong assumption: noise depends on gates but not their order
- Single aggregate error metric (no specificity)

- XEB was designed for the quantum supremacy regime: the *idea* is that if a noisy circuit produces bitstrings whose ideal probabilities are higher than chance, it's computing something
- **Porter–Thomas distribution**: for random circuits, the ideal output probabilities follow an exponential distribution — this is essential for XEB to work, as it provides the variance structure needed
- XEB's blindness to I/Z errors is a significant limitation: in the computational basis, a Z error changes only the phase and not the measurement outcome
- RCS benchmarking mirrors RB's structure: random circuits play the role of random Cliffords, and the exponential decay is analogous

4 Deterministic Benchmarking (DB) – Novel Contribution

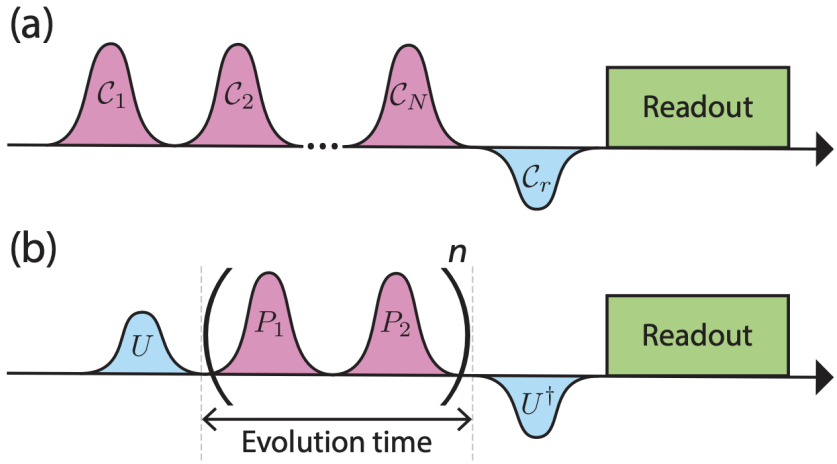

Figure 5: RB vs. DB: (a) RB uses N random Cliffords $\mathcal{C}_i$ + recovery $\mathcal{C}_r$; (b) DB uses state prep U, n repetitions of fixed pulse-pair P_1P_2, un-prep $U^\dagger$.

4.1 Qubit Model

- Transmon qubit modeled as **driven two-level system** in the drive frame (rotating wave approximation)
- Hamiltonian generating gates $R_x(\theta)$, $R_y(\theta)$:

$$H_\alpha = (\varepsilon + \varepsilon_{\text{err}})\frac{\sigma_\alpha}{2} + \Delta_{\text{err}}\frac{\sigma_z}{2}, \qquad \alpha \in \{x, y\} \tag{6}$$

- **Rotation error**: $\delta\theta \equiv \varepsilon_{\text{err}}\, t_g$ (amplitude error × gate time)
- **Phase error**: $\delta\phi \equiv \Delta_{\text{err}}/\bar{\varepsilon} = \Delta_{\text{err}}\, t_g/\theta$ (detuning relative to drive)
- Δ_{err} arises from both detuning and ac-Stark shift from driving higher transitions

4.2 Four Key Parameters

While a general single-qubit CPTP map has 12 parameters, DB reduces to **4 key parameters**: $\{T_1, T_2, \delta\theta, \delta\phi\}$

Parameter	Type	Physical Origin	DB Sequence
T_1	Incoherent	Relaxation to equilibrium	Free evolution; $\|1\rangle$
T_2	Incoherent	Coherence time ($1/T_2 = 1/2T_1 + 1/T_\phi$)	XX; $\|+\rangle$
$\delta\theta$	Coherent	Rotation (amplitude) error	YY; $\|+\rangle$
$\delta\phi$	Coherent	Phase (detuning) error	$X\bar{X}$; $\|+\rangle$

Gate-pair sequences: $P_1P_2 \in \{XX, X\bar{X}, YY, Y\bar{Y}, \bar{Y}Y\}$, where $X \equiv R_x(\pi)$, $\bar{X} \equiv R_x(-\pi)$, etc. Without errors: $P_1P_2 = \pm I$.

4.3 DB Protocol

Universal fidelity expression: In all experiments, the fidelity is accurately described by:

$$\boxed{F(t_n) = \frac{1}{2}(1+a) + \frac{1}{2}(1-a)\, e^{-t_n/T_D}\cos(2\omega t_n)}$$

where $t_n = 2n\, t_g$, T_D = decay timescale (incoherent), ω = oscillation frequency (coherent), a = deviation from infinite-T limit.

Learning experiments (Steps 1–4):

1. $\{$free; $|1\rangle\}$: Standard T_1 measurement. $F = \frac{1}{2}(1+a) + \frac{1}{2}(1-a)e^{-t/T_1} \Rightarrow$ extract T_1
2. $\{XX;\ |+\rangle\}$: Exponential decay $\Rightarrow$ extract T_2 (analogous to echo-Ramsey). $T_\phi = 2T_1T_2/(2T_1 - T_2)$
3. $\{YY;\ |+\rangle\}$: Oscillations $\Rightarrow$ fit to find $\delta\theta$ where $\omega \approx \delta\theta/(2t_g)$
4. $\{X\bar{X};\ |+\rangle\}$: Oscillations $\Rightarrow$ fit to find $\delta\phi$ where $\omega = \delta\phi/t_g$

Test experiments (Steps 5–6):

5. $\{Y\bar{Y};\ |+\rangle\}$ and $\{\bar{Y}Y;\ |+\rangle\}$: Tests interplay of relaxation and gate operations; validates $\{T_1, T_2, \delta\theta, \delta\phi\}$; can characterize qubit temperature
6. Various (e.g., UR6 DD sequence): Further validation; UR6 suppresses coherent errors

Notation: $\{P_1P_2;\ |\psi\rangle\}$ means: prepare $|\psi\rangle$ from $|0\rangle$, apply $(P_1P_2)^n$, un-prepare $|\psi\rangle$, measure in σ_z basis.

4.4 Experimental Results

Device: Superconducting grounded Xmon transmon, dispersively coupled to CPW readout resonator. DRAG-calibrated pulses, cosine envelope, $t_g = 88$ ns (80 ns pulse + 8 ns padding).

$\omega_q/2\pi$	$\lvert\eta\rvert/2\pi$	$\chi_{qc}/2\pi$	$\omega_c/2\pi$	T_1	T_2^*
4.37 GHz	215 MHz	230 kHz	6.95 GHz	20 μs	32 μs

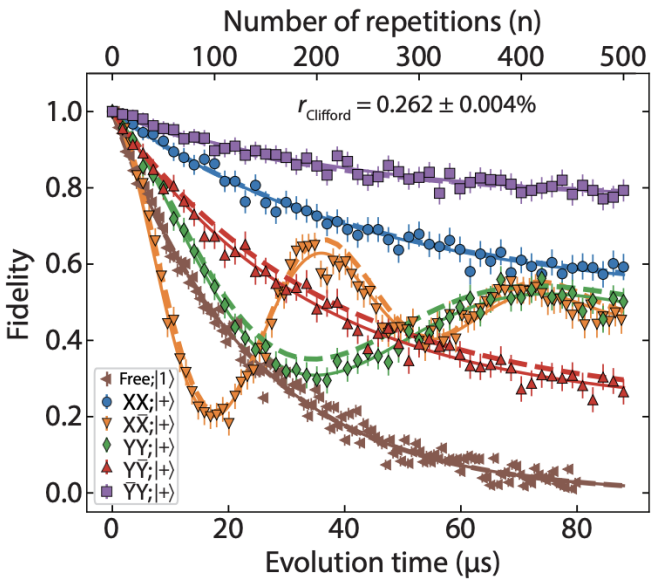

Figure 6: DB fidelities: symbols = experiment, solid = analytical fit [Eq. above], dashed = Lindblad simulation. Extracted: $T_1 = 23.36 \pm 0.40$ μs, $T_2 = 44.13 \pm 2.49$ μs, $\delta\theta = 0.398 \pm 0.004°$, $\delta\phi = 0.426 \pm 0.004°$.

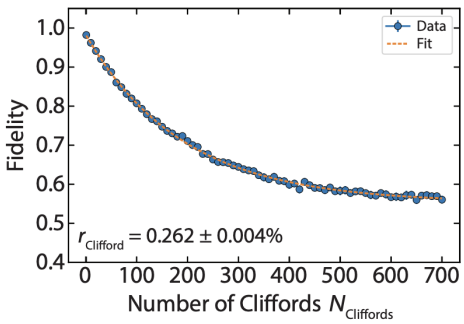

Figure 7: RB for same parameters: $r_C = 0.262 \pm 0.004\%$ over 30 random sequences with up to 700 Cliffords. RB misses the coherent oscillations and T_1/T_2 distinction.

Key observations:

- YY oscillates $\Rightarrow$ rotation errors; $X\bar{X}$ oscillates $\Rightarrow$ phase errors
- XX on $|+\rangle$ is insensitive to coherent errors: pure exponential decay $\Rightarrow T_2$
- Strong **asymmetry** between $Y\bar{Y}$ and $\bar{Y}Y$: $Y\bar{Y}$ confines state to southern (excited) hemisphere $\Rightarrow$ faster decay; $\bar{Y}Y$ stays in northern hemisphere
- RB gives $r_C = 0.262\%$ — a single number that misses all this structure

4.5 Closed-System (Coherent Error) Fidelities

Analytical results for initial state $|+\rangle$, $\theta_{\text{err}} \equiv \sqrt{(\varepsilon t_g + \delta\theta)^2 + (\pi\delta\phi)^2}$:

$$F_{YY}^{\text{clsd}}(n) = \cos^2(n\,\theta_{\text{err}}) \xrightarrow{\text{small errors}} \cos^2(n\,\delta\theta) \qquad \text{dominated by } \delta\theta \tag{7}$$

$$F_{XX}^{\text{clsd}}(n) = 1 - \left(\frac{\pi\delta\phi\sin(n\theta_{\text{err}})}{\theta_{\text{err}}}\right)^2 \xrightarrow{\text{small errors}} 1 \qquad \text{insensitive to errors} \tag{8}$$

$$F_{X\bar{X}}^{\text{clsd}}(n) = \cos^2(n\,\phi_{\text{err}}) \xrightarrow{\text{small errors}} \cos^2\left(\tfrac{1}{2}n\delta\phi[1-\pi\delta\theta]\right) \qquad \text{dominated by } \delta\phi \tag{9}$$

4.6 Open System Model

Lindblad master equation: $\partial_t\rho = \mathcal{L}(\rho) = -i[H,\rho] + \sum_\alpha \frac{\gamma_\alpha}{2}(L_\alpha\rho L_\alpha^\dagger - \{L_\alpha^\dagger L_\alpha, \rho\})$

Lindblad operators: $L_1 = \sigma^- = |0\rangle\langle 1|$ ($\gamma_1 = 1/T_1$), $L_\phi = \sigma_z/\sqrt{2}$ ($\gamma_\phi = 1/T_\phi$).

Using fitted $\{T_1, T_2, \delta\theta, \delta\phi\}$ from Steps 1–4 $\Rightarrow$ Lindblad simulations predict Steps 5–6 accurately (dashed curves in Fig. 6).

4.7 Sensitivity to Coherent Errors

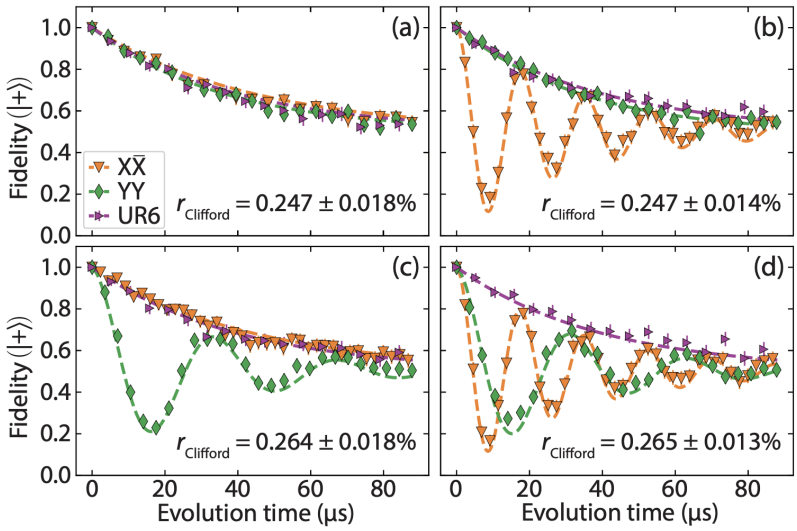

Figure 8: Sensitivity comparison DB vs. RB: (a) calibrated gates ($\delta\theta = \delta\phi = 0$), (b) phase error only, (c) rotation error only, (d) both errors. RB's r_C is essentially unchanged despite significant coherent errors visible in DB.

Critical finding: Small coherent errors ($\delta\phi \approx 0.9°$, $\delta\theta \approx 0.93°$) cause large, rapid oscillations in DB fidelity, while **RB's r_C changes by** $< 0.02\%$. This demonstrates RB's fundamental limitation.

UR6 sequence never exhibits oscillations $\Rightarrow$ confirms oscillations are from coherent errors (UR6 suppresses them by design).

Induced rotation error from detuning: $\delta\theta_{\text{ind}} \approx \frac{1}{2}\varphi\,\delta\phi^2$ (negligible for small $\delta\phi$).

4.8 T_1 Asymmetry and Temperature

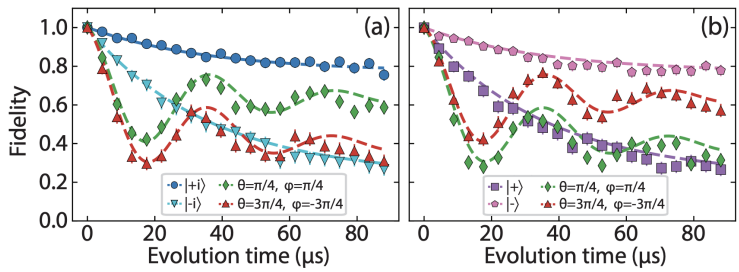

Figure 9: Asymmetric decay under (a) XX on $|\pm i\rangle$ and other states, (b) YY on $|\pm\rangle$ and other states. Orthogonal states show opposite asymmetry due to hemisphere confinement.

- Transmon operates at $\omega_{01} \in [3\text{–}6]$ GHz, $T \sim 10$ mK (~ 0.2 GHz) $\Rightarrow$ ground state $\approx$ thermal state

- $Y\bar{Y}$ confines state to **southern hemisphere** (excited) $\Rightarrow$ susceptible to T_1 relaxation $\Rightarrow$ faster decay
- $\bar{Y}Y$ confines to **northern hemisphere** (ground) $\Rightarrow$ less relaxation $\Rightarrow$ slower decay, higher saturation
- This asymmetry enables **temperature calibration**: $k_B T = \omega_{01}\left[\ln\left(\frac{a_{T=0}+a}{a_{T=0}-a}\right)\right]^{-1}$
- Choose pulse sequences with larger $|a_{T=0}|$ (e.g., $Y\bar{Y}$) for more sensitive temperature measurement

4.9 Analytical Derivations

4.9.1 Bloch Vector Formulation

$\rho = \frac{1}{2}(I + v_x\sigma_x + v_y\sigma_y + v_z\sigma_z)$, $\quad H = \frac{1}{2}(h_x\sigma_x + h_y\sigma_y + h_z\sigma_z)$

Dissipator: $R = \mathrm{diag}(-\gamma_2, -\gamma_2, -\gamma_1)$, $c = (0, 0, \eta\gamma_1)$, with $\gamma_2 = \frac{1}{2}\gamma_1 + \gamma_\phi$ and $\eta = \frac{1-e^{-\beta\omega_{01}}}{1+e^{-\beta\omega_{01}}} \in [0, 1]$.

Bloch equation: $\dot{v} = h \times v + Rv + c$. $\quad$ Fidelity with $|+\rangle$: $F(t) = \frac{1}{2}(1 + v_x(t))$.

4.9.2 YY Sequence ($\Delta_{\mathrm{err}} = 0$)

For square YY pulse, $h = (0, \varepsilon_{\mathrm{tot}}, 0)$. Solving $\dot{v} = Gv + c$ with $v(0) = (1, 0, 0)$:

$$v_\infty = \eta\frac{\varepsilon_{\mathrm{tot}}^2\gamma_1}{\varepsilon_{\mathrm{tot}}^2 + \gamma_1\gamma_2}, \quad \gamma_* = \frac{1}{2}(\gamma_1 + \gamma_2), \quad \omega_* = \sqrt{\varepsilon_{\mathrm{tot}}^2 - \frac{1}{4}(\gamma_1 - \gamma_2)^2} \tag{10}$$

In the regime $\varepsilon \gg \varepsilon_{\mathrm{err}}, \gamma_1, \gamma_2$: $v_\infty \approx \eta\gamma_1/\varepsilon$, $\omega_* \approx \varepsilon_{\mathrm{tot}}$:

$$F_{\delta\phi=0}(t_n) = \frac{1}{2}(1 + v_\infty) + \frac{1}{2}(1 - v_\infty)e^{-\gamma_* t_n}\cos(2n\delta\theta) \tag{11}$$

This matches the phenomenological form exactly: $a = v_\infty$, $T_D = 1/\gamma_*$, $2\omega t_g = \delta\theta$.
For $\Delta_{\mathrm{err}} \neq 0$: perturbative correction $|F_{YY}(n) - F_{\delta\phi=0}(n)| = O(\gamma_1/\varepsilon) = O(\gamma_1 t_g)$.

4.9.3 XX Sequence

For the simplified model $\gamma_1 = \gamma_2 = \gamma$ (i.e., $T_\phi = 2T_1$), the Bloch vector after n XX pulse-pairs satisfies a recurrence $v(2(n+1)t_g) = Av(2nt_g) - \zeta$. Under first-order approximations:

$$F(t_n) = \frac{1}{2} + \frac{1}{2}e^{-\gamma t_n}\cos(2n\vartheta), \quad \vartheta = \tan^{-1}\left(2\sin\frac{\omega' t_g}{2} \cdot \frac{\Delta_{\mathrm{err}}}{\omega'}\right) \tag{12}$$

where $\omega' = \sqrt{\varepsilon_{\mathrm{tot}}^2 + \Delta_{\mathrm{err}}^2}$, and the oscillation frequency matches the closed-system result.

4.10 Leakage Diagnostics

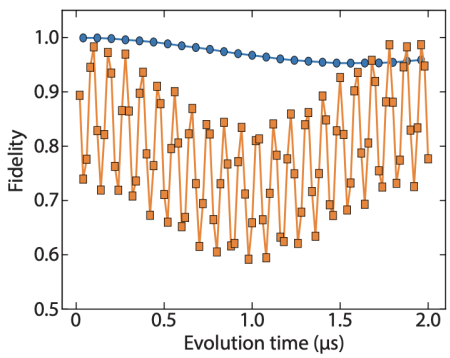

Figure 10: Simulated fidelity of $|+\rangle$ under XX with leakage to a state -150 MHz detuned. 20 ns pulses (blue) show minimal leakage; 10 ns pulses (orange) show significant oscillations. DRAG can remove these.

- XX on $|+\rangle$ has *no oscillations* absent leakage $\Rightarrow$ any oscillations in XX are a **direct leakage diagnostic**
- Faster gates (larger bandwidth) $\Rightarrow$ more leakage $\Rightarrow$ larger/faster oscillations
- DRAG pulse optimization can suppress leakage; residual leakage detectable via DB
- Useful when leakage state is unknown (direct population measurement impossible)

4.11 DB Summary

- **Only 4 experiments** needed (vs. 80+ for GST, D^4 for QPT)
- **SPAM-resilient** (exponential decay separates SPAM)
- Detects **coherent errors invisible to RB**
- Distinguishes T_1 from T_2
- Enables **temperature calibration**
- Enables **leakage diagnostics**
- Simple analytical model; Lindblad validation
- Platform-independent (validated on transmon)

- **Currently limited to single-qubit gates**
- 4 parameters (not full 12-parameter CPTP characterization)
- Assumes Markovian noise model
- Extension to 2-qubit gates is an open problem
- Not yet widely adopted/tested

- DB's philosophy: *target the right physics* rather than brute-force characterize everything
- The choice of XX, YY, $X\bar{X}$ sequences is not arbitrary — each isolates one specific error type by exploiting the qubit's Bloch sphere geometry
- XX on $|+\rangle$: the state remains at $|+\rangle$ under ideal noiseless evolution $\Rightarrow$ any decay is purely incoherent $\Rightarrow$ clean T_2 measurement
- The $Y\bar{Y}$ vs. $\bar{Y}Y$ asymmetry is deeply physical: it reflects the asymmetry between absorption and emission at finite temperature, modulated by where the pulse sequence keeps the state on the Bloch sphere
- DB's connection to **dynamical decoupling**: same pulse sequences, but DD tries to *suppress* noise while DB tries to *characterize* it — a clever inversion of purpose
- The phenomenological fidelity $F(t_n) = \frac{1}{2}(1+a)+\frac{1}{2}(1-a)e^{-t_n/T_D}\cos(2\omega t_n)$ naturally emerges from the Lindblad equation, validating the 4-parameter model
- **Noise bias** revealed by DB (asymmetry in relaxation during gates) could inform compilation strategies that minimize time spent near excited states

5 Conclusions and Method Comparison

Method	Params	SPAM	Coherent	Scaling	Cost	Scope	Key Use Case
RB	1	robust	no	good	low	gate	Quick fidelity check
IRB	1	robust	no	good	low	gate	Specific gate fidelity
SQPT	D^4-D^2	sensitive	yes	poor	high	gate	Full characterization
AAPT	D^4-D^2	sensitive	yes	poor	high	gate	Full char. w/ ancilla
DCQD	D^4-D^2	partial	yes	moderate	mod.	gate	Efficient full char.
GST	D^4-D^2	robust	yes	poor	v. high	gate set	Gold standard + SPAM
DFE	1	sensitive	yes	moderate	mod.	gate/circ.	Fidelity estimation
PFE	1	partial	yes	moderate	mod.	circuit	End-to-end fidelity
XEB	1	robust	partial	good	mod.	circuit	Supremacy; noise rate
DB	4	robust	yes	1Q only	v. low	gate	Coherent + incoherent

Table 1: Comparison of benchmarking methods. Color coding: green = advantage, red = disadvantage, orange = partial.

Key takeaways:

- **No universal winner**: method choice depends on goal (calibration, performance prediction, processor quality)
- **RB** = cheapest but least informative; **GST/QPT** = most informative but most expensive
- **DB** fills a gap: more informative than RB, much cheaper than GST, detects coherent errors
- **XEB** is unique in scaling to large systems but requires classical simulability
- Noise-reduction strategies (error mitigation, QEC) *require* accurate noise characterization ⇒ benchmarking is a prerequisite for fault tolerance
- Open problem: extending DB to 2-qubit gates with minimal parameter set

- **Expert strategy**: Use RB for quick monitoring, DB for detailed single-qubit calibration, GST when you need the full picture and can afford the cost, XEB for circuit-level assessment
- **Coherent errors are the silent killer**: they accumulate quadratically, are invisible to RB, and can dominate in well-engineered devices where incoherent errors have been minimized
- **Benchmarking ≠ characterization**: benchmarking gives metrics, characterization gives the error map. Choose based on whether you need to *compare* or *fix* the gate
- As quantum chemistry pushes toward chemical accuracy, the demands on gate fidelity will increase dramatically — benchmarking methods must evolve correspondingly